

Hardik Gupta

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EDUCATION

PES University, *B.Tech in Electronic and Communication Engineering*

Aug 2023 – Present — Bangalore, India

Lotus Valley International School, *Class 12th- CBSE*

Apr 2022 – Mar 2023 — Noida, India

Lotus Valley International School, *Class 10th- CBSE*

Apr 2020 – Mar 2021 — Noida, India

CERTIFICATION

AI & ML with Drone Technology - IIT Hyderabad (Tihan)

Jun 2025 – Feb 2026

- Hands-on training in **ML**, **Deep Learning**, and AI deployment with academic exposure from **IIT Hyderabad**.
- Built and evaluated **regression**, **classification**, **clustering**, and **neural network** models using Python libraries.
- Gained practical understanding of the building blocks and working principles of drones, including core components.

EXPERIENCE

Bharat Space Education Research Centre, *New Delhi*

Dec 2025 – Jan 2026

Winter Intern

- Modeled aircraft stability and performance using **MATLAB (AID toolbox)** and **XFLR5**, identifying key design parameters for optimal flight behavior.

ColorHome Lighting, *Noida*

Jul 2025 – Aug 2025

Summer Intern

- Analyzed the end-to-end manufacturing lifecycle of 9W LED bulbs, from component sourcing to final dispatch.
- Gained hands-on experience in **SMT operations**, **quality assurance**, and **electrical safety testing**.

PROJECTS

Vocab Ledger – Native Android Vocabulary App ▶

Jul 2026

- Architected an offline-first native Android app using **Kotlin**, **Jetpack Compose**, **Room DB**, and **Firebase**, featuring B-Tree database indexing and cloud sync; published in Closed Testing on **Google Play Store**.
- Optimized release binary size by **44%** (from 19.7 MB down to 11.0 MB) by configuring custom **R8/ProGuard** minification and resource shrinking rules.

Automated PCB Defect Detection System 📷

Apr 2026

- Trained and benchmarked **YOLOv5s** and **YOLOv8s** on 693 PCB images to detect and classify 6 manufacturing defect types, achieving **97.4% mAP@0.5** and **95.2% recall**.
- Built a 4-level severity analysis system that evaluates defect area, position, and confidence to deliver automated per-board PASS/FAIL verdicts with repair recommendations.

Automatic Audio Synthesis for Silent Videos 📺

Mar 2026

- Implemented the AutoFoley Frame-Relation Network (**ResNet-50 + Multiscale TRN**) to predict and synthesize audio for silent videos across 10, 12, and 15 sound-class configurations using VGGSound and AVSync15 datasets.
- Built an end-to-end pipeline covering frame extraction, per-class average spectrogram computation, temporal relation-based classification, and **Griffin-Lim audio reconstruction** from predicted spectrograms.

Federated AI for Privacy-Preserving Alzheimer's Detection 🧠

Aug 2025 – Present

- **Designing a secure Federated Learning pipeline** for Alzheimer's MRI classification using **VGG19/ResNet50**, enabling decentralized model training on private data.
- **Integrating Grad-CAM and SHAP** to generate visual heatmaps and feature importance scores, improving the interpretability of the model's predictions.

SKILLS

Languages & Mobile: Python, C, Kotlin, Assembly, Jetpack Compose, Room DB, Firebase

Machine Learning & CV: Scikit-learn, TensorFlow, PyTorch, OpenCV, YOLOv5/v8, NumPy, Pandas, Matplotlib, Seaborn

Tools & Frameworks: Git, Jupyter, VS Code, MATLAB, Google Colab, Ripes, Arduino IDE, LaTeX